

LSM - W6 Assignment

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(1)
(i)

Proove μ^* is unbiased estimator of μ under condition $\sum a_i = 1$

μ^* is unbiased estimator of μ if $E(\mu^*) = \mu$

Given

$$\mu^* = \sum a_i X_i$$

$$\Rightarrow E(\mu^*) = E(\sum a_i X_i)$$

Since E is linear operator

$$E(\sum a_i X_i) = \sum E(a_i X_i) = \sum_{i=1}^n a_i E(X_i)$$

$$E(\mu^*) = \sum_{i=1}^n a_i (\mu) = \mu \sum_{i=1}^n a_i$$

Since $(\sum a_i = 1, \text{ given})$

$$E(\mu^*) = \mu, \text{ hence } \mu^* \text{ is}$$

unbiased estimator.

(ii) BLUE of form μ^*

When $\text{Var}(\mu^*)$ is minimum
then its BLUE

Now,

$$\text{Var}(\mu^*) = \text{Var}\left(\sum_{i=1}^n a_i x_i\right)$$

Since $x_1, \dots, x_n$ are random i.i.d samples

$$\text{Var}(\mu^*) = \sum_{i=1}^n \text{Var}(a_i x_i)$$

$$= \sum_{i=1}^n a_i^2 \text{Var}(x_i)$$

$$= \sum_{i=1}^n a_i^2 \sigma^2 = \sigma^2 \sum_{i=1}^n a_i^2$$

Now,

We want to minimise $\sum a_i^2$

given $\sum a_i = 1$

for Best Linear Unbiased Estimator

$$L(a_1, a_2, \dots, a_n, \lambda) = \sum a_i^2 - \lambda(\sum a_i - 1)$$

$$\frac{\partial L}{\partial a_i} = 2a_i - \lambda = 0 \Rightarrow a_i^* = \lambda/2$$

$$\Rightarrow a_1 = a_2 = a_3 = \dots = a_n = a^* = \lambda/2$$

Given $\sum a_i^0 = 1$

$$\Rightarrow n \cdot a^* = 1$$

$$\Rightarrow a^* = 1/n$$

Therefore $a_i^* = \frac{1}{n}$

Hence $\mu^* = \sum_{i=1}^n a_i^0 x_i = \sum_{i=1}^n \frac{1}{n} x_i$

$$= \frac{1}{n} \sum x_i = \bar{x}$$

2)

Given $\underset{\sim}{L}^T \underset{\sim}{y}$ is unbiased estimate of $\underset{\sim}{P}^T \underset{\sim}{\beta}$ & $\underset{\sim}{L}, \underset{\sim}{X}, \underset{\sim}{\beta}$ are not random variable

$$\Rightarrow E(\underset{\sim}{L}^T \underset{\sim}{y}) = \underset{\sim}{P}^T \underset{\sim}{\beta}$$

We put $\underset{\sim}{y} = \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\epsilon}$

$$E(\underset{\sim}{L}^T (\underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\epsilon})) = \underset{\sim}{P}^T \underset{\sim}{\beta}$$

$$= E(\underset{\sim}{L}^T \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{L}^T \underset{\sim}{\epsilon}) = \underset{\sim}{P}^T \underset{\sim}{\beta}$$

$$= E(\underset{\sim}{L}^T \underset{\sim}{X} \underset{\sim}{\beta}) + E(\underset{\sim}{L}^T \underset{\sim}{\epsilon}) = \underset{\sim}{P}^T \underset{\sim}{\beta}$$

$$= \underset{\sim}{L}^T \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{L}^T E(\underset{\sim}{\epsilon})^0 = \underset{\sim}{P}^T \underset{\sim}{\beta}$$

$$\Rightarrow \underset{\sim}{L}^T \underset{\sim}{X} \underset{\sim}{\beta} = \underset{\sim}{P}^T \underset{\sim}{\beta} \Rightarrow (\underset{\sim}{L}^T \underset{\sim}{X} - \underset{\sim}{P}^T) \underset{\sim}{\beta} = 0$$

Since this needs to hold true for

any β .

$$\Rightarrow \mathbf{X}^T \mathbf{X} - \mathbf{P}^T = 0$$

$$\Rightarrow \mathbf{X}^T \mathbf{X} = \mathbf{P}^T$$

$$\text{or } \boxed{\mathbf{X}^T \mathbf{1} = \mathbf{p}}$$

(3) $\rightarrow \mathbf{P}^T \beta$ is estimable if there exists a linear estimator $\hat{\mathbf{y}} = \mathbf{X}^T \mathbf{y}$ such that $\mathbb{E}(\hat{\mathbf{y}}) = \mathbf{P}^T \beta$

To prove:

$\mathbf{P}^T \beta$ is estimable iff $\mathbf{p}$ is orthogonal to Null Space ($\mathbf{X}$).

in last ques (Q2) we proved $\hat{\mathbf{y}} = \mathbf{X}^T \mathbf{y}$ is unbiased estimator of $\mathbf{P}^T \beta$ if $\mathbf{p} = \mathbf{X}^T \mathbf{1}$ or $\mathbf{p}$ is in column space of $\mathbf{X}^T$

$$\mathbf{p} \in \text{col}(\mathbf{X}^T) \Leftrightarrow \mathbf{p} \text{ is orthogonal to } \text{Null}(\mathbf{X})$$

(2) Estimable $\Rightarrow$ orthogonal

Assume $\mathbf{P}^T \beta$ is estimable

$$\Rightarrow p = X^T l$$

Now we need to show p is orthogonal to $N(X)$

Let $\underline{v}$ be any arbitrary vector in $N(X) \Rightarrow X \cdot \underline{v} = 0$

For p to be orthogonal to $N(X)$

$$\Rightarrow \underline{p} \cdot \underline{v} = 0 \Rightarrow \underline{p}^T \underline{v} = 0$$

$\underline{p} \cdot \underline{v}$ must be 0 or $\underline{p}^T \underline{v} = 0$

Now,

$$\underline{p}^T \underline{v} = (X^T l)^T \underline{v} = l^T X \underline{v}$$

but $X \underline{v} = 0 \Rightarrow l^T X \underline{v} = 0$

$$\Rightarrow \underline{p}^T \underline{v} = 0 \Rightarrow p \perp N(X)$$

(2) Orthogonal $\Rightarrow$ estimable.

We know already that $N(X) \perp C(X^T)$

Given p is orthogonal ~~to $N(X)$~~

$$\Rightarrow p \perp N(X)$$

$$\text{or } p \in C(X^T)$$

$$p = X^T a$$

Now

$$E(a^T \underline{y}) = E(a^T X \underline{\beta}) + E(a^T \underline{\epsilon}) = a^T X \underline{\beta}$$

Now,

$$E(\underline{a}^T \underline{y}) = \underline{a}^T \underline{x} \underline{\beta} = (\underline{x}^T \underline{a})^T \underline{\beta}$$

$$\text{but } \underline{p} = \underline{x}^T \underline{a}$$

$$\Rightarrow E(\underline{a}^T \underline{y}) = \underline{p}^T \underline{\beta}$$

Hence if $\underline{p} \in \text{Col}(\underline{X}^T)$ or
 $\underline{p}$ is orthogonal to $N(\underline{X})$

then $\underline{p}^T \underline{\beta}$ is estimable

$$\boxed{\underline{p}^T \underline{\beta} \text{ is estimable} \Leftrightarrow \underline{p} \perp N(\underline{X})}$$

(4) Given $\underline{L}^T \underline{y}$ & $\underline{\lambda}^T \underline{y}$ are both
Linear zero estimators
 $\Rightarrow E(\underline{L}^T \underline{y}) = E(\underline{\lambda}^T \underline{y}) = 0$

$$(i) \underline{L}^T \underline{y} - \underline{\lambda}^T \underline{y} = (\underline{L}^T - \underline{\lambda}^T) \underline{y} = (\underline{L} - \underline{\lambda})^T \underline{y}$$

Since $(\underline{L} - \underline{\lambda})$ is a vector in $\mathbb{R}^n$
Hence $\underline{L}^T \underline{y} - \underline{\lambda}^T \underline{y}$ is Linear
estimator

$$\text{Now, } E(\underline{L}^T \underline{y} - \underline{\lambda}^T \underline{y}) = E(\underline{L}^T \underline{y}) - E(\underline{\lambda}^T \underline{y})$$

$\Rightarrow (\underline{L}^T \underline{y} - \underline{\lambda}^T \underline{y})$ is a Linear zero est.

(ii)
$$\mathcal{L}^T y + \lambda^T y = (\mathcal{L} + \lambda)^T y$$

Since: $\mathcal{L}, \lambda \in \mathbb{R}^n \Rightarrow (\mathcal{L} + \lambda) \in \mathbb{R}^n$

Hence $\mathcal{L}^T y + \lambda^T y$ is linear

Now,

$$\mathbb{E}(\mathcal{L}^T y + \lambda^T y) = \mathbb{E}(\mathcal{L}^T y) + \mathbb{E}(\lambda^T y) = 0 + 0 = 0$$

$\Rightarrow \mathcal{L}^T y + \lambda^T y$ is a linear Zero estimator

(iii)
$$\mathcal{L}^T y \times \lambda^T y = \left(\sum \mathcal{L}_i y_i \right) \times \left(\sum \lambda_j y_j \right)$$

When expanded it will contain terms like $\mathcal{L}_i \lambda_j y_i y_j$ & $\mathcal{L}_i \lambda_j y_i^2$ which are quadratic but not linear.

Hence $\mathcal{L}^T y \times \lambda^T y$ is not linear zero estimator.

(Ans)

(i) $\mathcal{L}^T y - \lambda^T y$ is Linear Zero Est.

(ii) $\mathcal{L}^T y + \lambda^T y$ " " " "

(iii) $\mathcal{L}^T y \times \lambda^T y$ is not Linear Zero Est.